Experiment 05: Requirement Prioritization Using the MoSCoW Method

1. Aim

To categorize Library Management System requirements using the MoSCoW method and prioritize them based on user impact, feasibility, time, budget, and resources.

2. Tools Used

- Microsoft Excel or Google Sheets
- MoSCoW prioritization method
- Impact and feasibility scoring system

3. Procedure

1. List all 10 requirements of the Library Management System in Excel.
2. Categorize each requirement as Must-Have, Should-Have, Could-Have, or Won't-Have.
3. Assign an impact score from 1 to 5 based on importance to users.
4. Assign a feasibility score from 1 to 5 based on implementation difficulty and available resources.
5. Calculate the priority score by multiplying the impact score by the feasibility score.
6. Provide justification for each requirement's category.
7. Review the scores and categories to determine which features should be developed first.

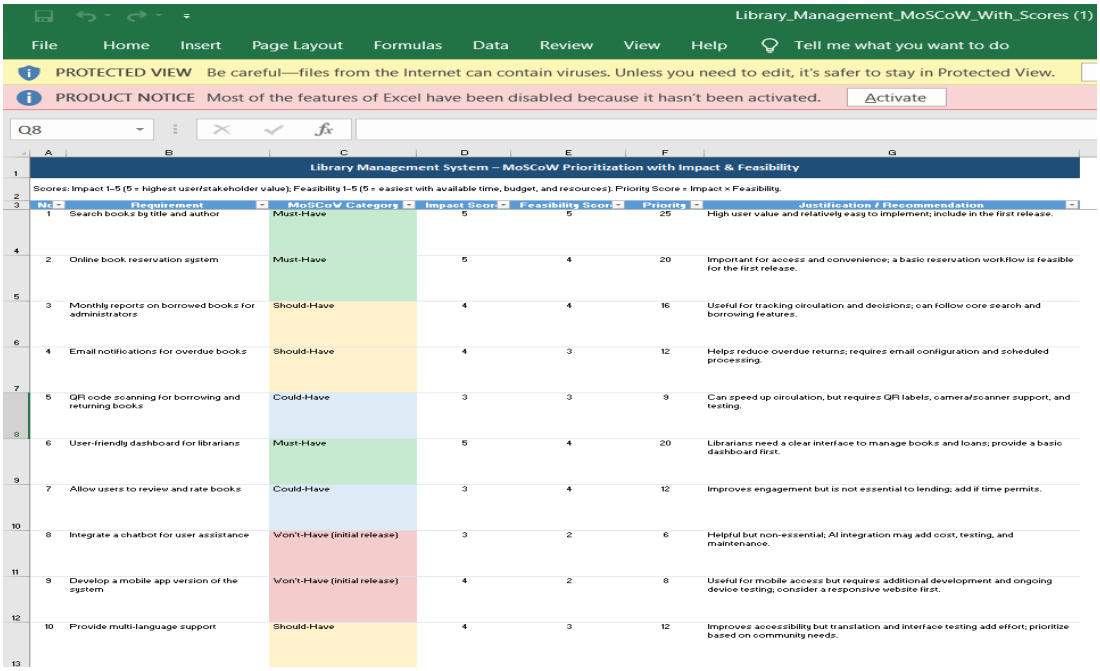
4. Algorithm

1. Start.
2. Read the list of requirements.
3. Evaluate the impact and feasibility of each requirement.
4. Assign a MoSCoW category.
5. Calculate Priority Score = Impact Score × Feasibility Score.
6. Record the justification for each decision.
7. Display the prioritized requirements.
8. Stop.

5. Result

The Library Management System requirements were categorized using the MoSCoW method, and impact, feasibility, and priority scores were assigned to help determine the order of implementation.

6.Output



Library_Management_MoSCoW_With_Scores (1)

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PROTECTED VIEW Be careful—files from the Internet can contain viruses. Unless you need to edit, it's safer to stay in Protected View.

PRODUCT NOTICE Most of the features of Excel have been disabled because it hasn't been activated. Activate

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Library Management System – MoSCoW Prioritization with Impact & Feasibility

Scores: Impact 1–5 (5 = highest user/stakeholder value); Feasibility 1–5 (5 = easiest with available time, budget, and resources). Priority Score = Impact × Feasibility.

No	Requirement	MoSCoW Category	Impact Score	Feasibility Score	Priority	Justification / Recommendation
1	Search books by title and author	Must-Have	5	5	25	High user value and relatively easy to implement; include in the first release.
2	Online book reservation system	Must-Have	5	4	20	Important for access and convenience; a basic reservation workflow is feasible for the first release.
3	Monthly reports on borrowed books for administrators	Should-Have	4	4	16	Useful for tracking circulation and decisions; can follow core search and borrowing features.
4	Email notifications for overdue books	Should-Have	4	3	12	Helps reduce overdue returns; requires email configuration and scheduled processing.
5	QR code scanning for borrowing and returning books	Could-Have	3	3	9	Can speed up circulation, but requires QR labels, camera/scanner support, and testing.
6	User-friendly dashboard for librarians	Must-Have	5	4	20	Librarians need a clear interface to manage books and loans; provide a basic dashboard first.
7	Allow users to review and rate books	Could-Have	3	4	12	Improves engagement but is not essential to lending; add if time permits.
8	Integrate a chatbot for user assistance	Won't-Have (initial release)	3	2	6	Helpful but non-essential; AI integration may add cost, testing, and maintenance.
9	Develop a mobile app version of the system	Won't-Have (initial release)	4	2	8	Useful for mobile access but requires additional development and ongoing device testing; consider a responsive website first.
10	Provide multi-language support	Should-Have	4	3	12	Improves accessibility but translation and interface testing add effort; prioritize based on community needs.